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ONLINE AUTHORIZING OF ROBOT AUTONOMY APPLICATIONS

Abstract

A method for online authoring of robot autonomy applications includes receiving sensor data of an environment about a robot while the robot traverses through the environment. The method also includes generating an environmental map representative of the environment about the robot based on the received sensor data. While generating the environmental map, the method includes localizing a current position of the robot within the environmental map and, at each corresponding target location of one or more target locations within the environment, recording a respective action for the robot to perform. The method also includes generating a behavior tree for navigating the robot to each corresponding target location and controlling the robot to perform the respective action at each corresponding target location within the environment during a future mission when the current position of the robot within the environmental map reaches the corresponding target location.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This U.S. Patent Application is a continuation of, and claims priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 18/466,535, filed on Sep. 13, 2023, which is a continuation of, and claims priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/884,954, filed on May 27, 2020, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application No. 63/009,119, filed on Apr. 13, 2020, each of which is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to online authoring of robot autonomy applications.

BACKGROUND

[0003] Robotic devices are increasingly being used to navigate constrained environments to perform a variety of tasks or functions. These robotic devices often need to navigate through these constrained environments without contacting the obstacles or becoming stuck or trapped. Typically, robots perform simultaneous localization and mapping (SLAM) to construct an environmental map of the robot's surroundings (including obstacles). After generation of the map, a user may interact with the generated map to set parameters for a future mission.

SUMMARY

[0004] One aspect of the disclosure provides a method for online authoring of robot autonomy applications. The method includes receiving, at data processing hardware of a robot, from at least one sensor in communication with the data processing hardware, sensor data of an environment about the robot while the robot traverses through the environment. The method also includes generating, by the data processing hardware, an environmental map representative of the environment about the robot based on the received sensor data of the environment about the robot while the robot traverses through the environment. While generating the environmental map, the method includes localizing, by the data processing hardware, a current position of the robot within the environmental map. The method also includes, at each corresponding target location of one or more target locations within the environment about the robot, recording, by the data processing hardware, a respective action for the robot to perform. The method also includes generating, by the data processing hardware, a behavior tree for navigating the robot to each corresponding target location of the one or more target locations and controlling the robot to perform the respective action at each corresponding target location of the one or more target locations within the environment during a future mission when the current position of the robot within the environmental map reaches the corresponding target location.

[0005] Implementations of the disclosure may include one or more of the following optional features. In some implementations, the respective action recorded for the robot to perform at the corresponding target location of at least one of the one or more target locations is selected from a playbook of actions associated with the corresponding target location. The playbook of actions associated with the corresponding target location may include a collection of prebuilt behavior tree

portions associated with the corresponding target location, and each prebuilt behavior tree portion may be associated with a different action for the robot to perform at the corresponding target location. Generating the behavior tree may include combining two or more prebuilt behavior tree portions, where each prebuilt behavior tree portion of the combined two or more prebuilt behavior tree portions is associated with a different corresponding target location and selected from the corresponding collection of prebuilt behavior tree portions associated with the corresponding target location.

[0006] In some examples, the method includes, prior to recording the respective action for the robot to perform, receiving, at the data processing hardware, from a user device in communication with the data processing hardware, an action recording request when the current position of the robot within the environmental map reaches the corresponding target location. The action recording request requests the data processing hardware to record the respective action for the robot to perform at the corresponding target location. The user device may be configured to display a graphical user interface on a screen of the user device. The graphical user interface has an action selection window presenting one or more available actions for the robot to perform. The user device may also receive, in the action selection window, the respective action for the robot to perform at each corresponding target location of the one or more target locations within the environment.

[0007] Optionally, in response to receiving the action recording request, the method includes obtaining, by the data processing hardware, additional information relevant to the respective action. Recording the respective action for the robot to perform at the corresponding target location may include recording the additional information relevant to the respective action for the robot to perform at the corresponding target location.

[0008] Recording the respective action for the robot to perform at the corresponding target location may include associating the current position of the robot within the environmental map with the respective action for the robot to perform at the corresponding target location and storing, in memory hardware in communication with the data processing hardware, the association of the current position of the robot within the environmental map with the respective action for the robot to perform at the corresponding target location.

[0009] In some implementations, recording the respective action for the robot to perform at the corresponding target location further includes obtaining additional information relevant to the respective action. The method may include, after generating the environmental map and the behavior tree, and while navigating the robot through the environment during the future mission, determining, by the data processing hardware, using the environmental map and the behavior tree, when the current position of the robot within the environmental map reaches one of the one or more target locations within the environment about the robot. When the current position of the robot within the environmental map reaches the one of the one or more target locations, the method may include controlling, by the data processing hardware, the robot to perform the respective action at the corresponding target location.

[0010] In some implementations, generating the environmental map includes receiving input from a user device directing the robot to move about the environment. As the robot moves about the environment, the method may include capturing the sensor data of the environment. Optionally, generating the behavior tree includes determining, for each respective action recorded for the robot to perform, a corresponding orientation for the robot.

[0011] In some examples, the method includes generating, by the data processing hardware, an updated environmental map representative of the environment about the robot based on new received sensor data of the environment about the robot while the robot traverses through the environment and generating, by the data processing hardware, a second behavior tree for controlling the robot to perform the respective action at each corresponding target location of the one or more target locations within the environment during the future mission when the current

position of the robot within the updated environmental map reaches the corresponding target location. The at least one sensor may include an image sensor including one or more of a stereo camera, a scanning light-detection and ranging (LIDAR) sensor, or a scanning laser-detection and ranging (LADAR) sensor.

[0012] Another aspect of the disclosure provides a system for online authoring of robot autonomy applications. The system includes data processing hardware and memory hardware in communication with the data processing hardware of a robot. The memory hardware stores instructions that when executed on the data processing hardware cause the data processing hardware to perform operations. The operations include receiving, from at least one sensor in communication with the data processing hardware, sensor data of an environment about the robot while the robot traverses through the environment. The operations also include generating an environmental map representative of the environment about the robot based on the received sensor data of the environment about the robot while the robot traverses through the environment. While generating the environmental map, the operations include localizing a current position of the robot within the environmental map. The operations also include, at each corresponding target location of one or more target locations within the environment about the robot, recording a respective action for the robot to perform. The operations also include generating a behavior tree for navigating the robot to each corresponding target location of the one or more target locations and controlling the robot to perform the respective action at each corresponding target location of the one or more target locations within the environment during a future mission when the current position of the robot within the environmental map reaches the corresponding target location.

[0013] This aspect may include one or more of the following optional features. In some implementations, the respective action recorded for the robot to perform at the corresponding target location of at least one of the one or more target locations is selected from a playbook of actions associated with the corresponding target location. The playbook of actions associated with the corresponding target location may include a collection of prebuilt behavior tree portions associated with the corresponding target location, and each prebuilt behavior tree portion may be associated with a different action for the robot to perform at the corresponding target location. Generating the behavior tree may include combining two or more prebuilt behavior tree portions, where each prebuilt behavior tree portion of the combined two or more prebuilt behavior tree portions is associated with a different corresponding target location and selected from the corresponding collection of prebuilt behavior tree portions associated with the corresponding target location.

[0014] In some examples, the operations include, prior to recording the respective action for the robot to perform, receiving, from a user device in communication with the data processing hardware, an action recording request when the current position of the robot within the environmental map reaches the corresponding target location. The action recording request requests the data processing hardware to record the respective action for the robot to perform at the corresponding target location. The user device may be configured to display a graphical user interface on a screen of the user device. The graphical user interface has an action selection window presenting one or more available actions for the robot to perform. The user device may also receive, in the action selection window, the respective action for the robot to perform at each corresponding target location of the one or more target locations within the environment.

[0015] Optionally, in response to receiving the action recording request, the operations include obtaining additional information relevant to the respective action. Recording the respective action for the robot to perform at the corresponding target location may include recording the additional information relevant to the respective action for the robot to perform at the corresponding target location.

[0016] Recording the respective action for the robot to perform at the corresponding target location may include associating the current position of the robot within the environmental map with the respective action for the robot to perform at the corresponding target location and storing, in the

memory hardware in communication with the data processing hardware, the association of the current position of the robot within the environmental map with the respective action for the robot to perform at the corresponding target location.

[0017] In some implementations, recording the respective action for the robot to perform at the corresponding target location further includes obtaining additional information relevant to the respective action. The operations may include, after generating the environmental map and the behavior tree, and while navigating the robot through the environment during the future mission, determining, using the environmental map and the behavior tree, when the current position of the robot within the environmental map reaches one of the one or more target locations within the environment about the robot. When the current position of the robot within the environmental map reaches the one of the one or more target locations, the operations may include controlling the robot to perform the respective action at the corresponding target location.

[0018] In some implementations, generating the environmental map includes receiving input from a user device directing the robot to move about the environment. As the robot moves about the environment, the operations may include capturing the sensor data of the environment. Optionally, generating the behavior tree includes determining, for each respective action recorded for the robot to perform, a corresponding orientation for the robot.

[0019] In some examples, the operations include generating an updated environmental map representative of the environment about the robot based on new received sensor data of the environment about the robot while the robot traverses through the environment and generating a second behavior tree for controlling the robot to perform the respective action at each corresponding target location of the one or more target locations within the environment during the future mission when the current position of the robot within the updated environmental map reaches the corresponding target location. The at least one sensor may include an image sensor including one or more of a stereo camera, a scanning light-detection and ranging (LIDAR) sensor, or a scanning laser-detection and ranging (LADAR) sensor.

[0020] The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0021] FIG. 1 is a schematic view of an example system for online authoring of robot autonomy applications.

[0022] FIGS. 2A and 2B are schematic views of exemplary environmental maps.

[0023] FIG. 3 is a schematic view of exemplary components of a user device.

[0024] FIG. 4 is a schematic view of a playbook of actions displayed on the user device of FIG. 3.

[0025] FIG. 5 is a schematic view of a traditional manually created behavior tree.

[0026] FIGS. 6A and 6B are schematic views of exemplary behavior trees automatically generated by components of the system of FIG. 1.

[0027] FIGS. 7A and 7B are schematic views of requests for additional information for an action displayed on the user device of FIG. 3.

[0028] FIG. 8 is a flowchart of an example arrangement of operations for a method of online authoring of robot autonomy applications.

[0029] FIG. 9 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

[0030] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0031] As robotic devices (also referred to as “robots”) become more prevalent, there is an increasing need to be able to quickly and easily map a space and set mission parameters for robots to perceive or interact with the environment about the robot autonomously. Today, this is accomplished by first maneuvering the robot around the environment to generate an environmental map. After the map is generated, the user interacts with the generated map to set mission parameters for the robot to complete autonomously. For example, the user may indicate a point on the environmental map that the robot should autonomously navigate to and perform an action (e.g., open a door). Thus, the user must separately map the environment and author behavior steps. Mapping the environment and then separately authoring behavior steps not only requires additional time, but also requires the user to be able to accurately interpret the environmental map in order to correctly author the intended actions. The nature of the generated environmental map (e.g., a topographical map) often makes this difficult. Users typically record the action by manually generating a behavior tree for the robot via an application programming interface (API) using a graphical user interface (GUI).

[0032] Implementations herein are directed toward an online authoring system that allows users of a robot to quickly map a space while simultaneously recording one or more actions to perform within that same space. The user may control or “drive” the robot through an environment to create a map and simultaneously localize that map so that the robot may track its location relative to the map. That is, the robot may perform simultaneous localization and mapping (SLAM). While performing SLAM (i.e., while the user navigates the robot through the environment), the online authoring system allows the user to also record one or more actions to perform during a future mission. The user may select an action from a playbook of actions and the robot may automatically generate a corresponding behavior tree during recording. The robot may combine prebuilt behavior tree portions or chunks to generate the behavior tree.

[0033] Referring to FIG. 1, a robot or robotic device **10** includes a body **11** with, for example, two or more legs **12** and executes an online authoring system **100** for enabling the robot **10** map the environment **8** while simultaneously recording one or more actions to perform within the environment. Each leg **12** is coupled to the body **11** and may have an upper portion **15** and a lower portion **16** separated by a leg joint **18**. The lower portion **16** of each leg **12** ends in a foot **19**. The foot **19** of each leg is optional and the terminal end of the lower portion of one or more of the leg **12** may be coupled to a wheel or the distal end of each leg **12** may directly contact the a ground surface **9**. The robot **10** has a vertical gravitational axis V_g along a direction of gravity, and a center of mass CM, which is a point where the weighted relative position of the distributed mass of the robot **10** sums to zero. The robot **10** further has a pose P based on the CM relative to the vertical gravitational axis V_g (i.e., the fixed reference frame with respect to gravity) to define a particular attitude or stance assumed by the robot **10**. The attitude of the robot **10** can be defined by an orientation or an angular position of the robot **10** in space. Movement by the legs **12** relative to the body **11** alters the pose P of the robot **10** (i.e., the combination of the position of the CM of the robot and the attitude or orientation of the robot **10**).

[0034] In some implementations, the robot **10** further includes one or more appendages, such as an articulated arm **20** disposed on the body **11** and configured to move relative to the body **11**. The articulated arm **20** may have five-degrees or more of freedom. Moreover, the articulated arm **20** may be interchangeably referred to as a manipulator arm or simply an appendage. In the example shown, the articulated arm **20** includes two portions **22**, **24** rotatable relative to one another and also the body **11**; however, the articulated arm **20** may include more or less portions without departing from the scope of the present disclosure. The first portion **22** may be separated from second portion **24** by an articulated arm joint **26**. An end effector **28**, which may be interchangeably referred to as a manipulator head **28**, may be coupled to a distal end of the second portion **24** of the articulated arm **20** and may include one or more actuators **29** for gripping/grasping objects.

[0035] The robot **10** also includes a vision system **30** with at least one imaging sensor or camera

31, each sensor or camera **31** capturing image data or sensor data **17** of the environment **8** surrounding the robot **10** with an angle of view **32** and within a field of view **34**. The vision system **30** may be configured to move the field of view **34** by adjusting the angle of view **32** or by panning and/or tilting (either independently or via the robot **10**) the camera **31** to move the field of view **34** in any direction. Alternatively, the vision system **30** may include multiple sensors or cameras **31** such that the vision system **30** captures a generally 360-degree field of view around the robot **10**. [0036] The camera(s) **31** of the vision system **30**, in some implementations, include one or more stereo cameras (e.g., one or more RGBD stereo cameras). In other examples, the vision system **30** includes one or more radar sensors such as a scanning light-detection and ranging (LIDAR) sensor, or a scanning laser-detection and ranging (LADAR) sensor, a light scanner, a time-of-flight sensor, or any other three-dimensional (3D) volumetric image sensor (or any such combination of sensors). In some implementations, the online authoring system **100** receives and processes the sensor data **17** and passes the processed sensor data **17** to a map generator and localizer (MGAL) **110**.

[0037] The vision system **30** provides image data or sensor data **17** derived from image data captured by the cameras or sensors **31** to data processing hardware **36** of the robot **10**. The data processing hardware **36** is in digital communication with memory hardware **38** and, in some implementations, may be a remote system. The remote system may be a single computer, multiple computers, or a distributed system (e.g., a cloud environment) having scalable/elastic computing resources and/or storage resources. An online authoring system **100** of the robot **10** executes on the data processing hardware **36**.

[0038] Using the processed sensor data **17**, the MGAL **110** generates an environmental map **114** representative of the environment **8** about the robot **10** while the robot traverses through the environment **8**. For example, the MGAL **110** receives input **44** from a user device **300** associated with a user **42** (i.e., the operator) to move about the environment **8**. As the robot **10** moves about the environment **8**, the vision system **14** captures sensor data **17** of the environment **8**. While generating the environmental map **114**, the MGAL **110** also localizes a current position of the robot **10** within the environmental map **114**. That is, in some implementations, the MGAL **110** performs SLAM using the sensor data **17**. The MGAL **110** may use any SLAM techniques, such as Kalman filters, Monte Carlo methods, self-membership techniques, and maximum a posteriori estimation.

[0039] While performing SLAM, an action recorder **120** of the online authoring system **100** may, at each corresponding target location **52** of one or more target locations **52** within the environment **8** about the robot **10**, record a respective action **420** (FIG. 4) for the robot **10** to perform (e.g., during a future mission). In the example shown, the target location **52** is in front of a door **50** and the action may be opening the door **50**. In this situation, the target location **52** is positioned such that when the robot **10** autonomously navigates to the target location **52** during a future mission, the robot **10** may use the articulated arm **20** and associated end effector **28** to reach for, and grasp, the door **50** to perform the action **420** (i.e., open the door).

[0040] The action recorder **120** receives the environmental map **114** from the MGAL **110**. In some examples, the action recorder **120** also receives processed sensor data **17** from the MGAL **110**. The action recorder **120** automatically generates a behavior tree **122** for navigating and controlling the robot **10** to perform the respective action **420** at each corresponding target location **52** of the one or more target locations **52** within the environment **8** during the future mission when the current position of the robot **10** within the environmental map **114** reaches the corresponding target location **52**. That is, the behavior tree **122** includes a series of instructions for the robot **10** to follow that allows the robot to navigate to each of the target locations **52** and perform the specified action **420**. The behavior tree **122** describes the switchings between a finite set of tasks in a modular fashion. The behavior tree **122** may include nodes **622** (FIGS. 6A and 6B) that describe simple tasks that, when combined, create a complex task associated with an action **420**.

[0041] FIGS. 2A and 2B include example user interfaces **216a**, **116b** depicting the MGAL **110** generating environmental maps **114a**, **114b** in response to user input **44** and the captured sensor

data **17**. The user interface includes a window displaying the generated maps **114a**, **114b** and a map option window **254** presenting user-selectable options to filter and otherwise manipulate data of the map **114**. For example, environmental map **114a** (FIG. 2A) includes waypoints **250**, **250a-n** placed by the robot **10** while traversing the environment **8** that loosely represent real space. This environmental map **114a** does not include any high level building information. Map options **255** allow the user **42** (via, for example, the user device **300**) to filter and otherwise manipulate the data of the environmental map **114a**. As another example, the environmental map **114b** (FIG. 2B) includes a blueprint **260** of the environment **8** overlain on the environmental map **114b**. This example also includes a texture **170** generated by the robot **10** from the sensor data **17** to increase context for the user **42** when creating and/or planning missions.

[0042] Changes in the environment (e.g., the addition or removal of obstacles) may outdate the current environment map **114**. That is, the robot **10** may no longer be able to successfully navigate the environment **8** using the outdated environmental map **114**. In this situation, the MGAL **110** may generate an updated environmental map **114** representative of the environment **8** about the robot **10** based on new received sensor data **17** of the environment **8** about the robot **10** while the robot traverses through the environment **8**. That is, the robot **10** may again traverse the environment **8** recording sensor data **17** to update the environmental map **114** to reflect any changes to the environment **8** since generating the last environmental map **114**. Optionally, when updating the environmental map **114**, the action recorder **120** may also simultaneously, and automatically generate a second/updated behavior tree **122** for navigating and/or controlling the robot **10** to perform the respective action **420** at each corresponding target location **52** of the one or more target locations **52** within the environment **8** during the future mission when the current position of the robot **10** within the updated environmental map **114** reaches the corresponding target location **52**. The second or updated behavior tree **122** thus updates the mission to include any environmental changes reflected in the updated environmental map **114**. Thus, the online authoring system **100** saves the user **42** time by allowing the reuse of old actions **420** with a new or updated environmental map **114**, as the user **42** does not need to re-record actions **420** after recording the actions **420** a first time while regenerating the environmental map **114**. In some examples, the online authoring system **100** reuses old actions **420** when locations in the updated environmental map **114** maintain the same names as equivalent locations within the old or original environmental map **114**. For example, one or more of the waypoints **250** in the environmental map **114** may each be associated with, or otherwise assigned/labeled, a specific name. The specific names associated with these waypoints **250** may transfer to equivalent waypoints in an updated version of the environmental map **114**. Maintaining the same name with the same or similar waypoints **250** between the original environmental map **114** and the updated environmental map **114** may allow the online authoring system **100** to transfer the action **420** to the updated environmental map **114** without requiring re-recording of the action **420**.

[0043] In some implementations, at least a portion of the online authoring system **100** executes on a remote device in communication with the robot **10**. For instance, the action recorder **120** may execute on a remote device to record actions **420** while the map generator and localizer **110** executes locally on the robot **10**. Optionally, the entire online authoring system **100** may execute on a remote device and the vision system **14** may transmit sensor data **17** to the remote device for map generation and action recording. In some examples, the remote device includes the user device **300** executing at least a portion of the online authoring system **100**. In additional examples, the remote device includes a remote computing device (e.g., a cloud-based server) in communication with the user device **300** and the robot **10**.

[0044] FIG. 3 shows an example user device **300** used by the user **42** for controlling/navigating the robot **10** around the environment **8**. The user device **300** includes a housing **310** with a screen **320** and a plurality of controls **330** situated around the screen **320**. As shown, the controls **330** are located on the front and top edge of the housing **310**, and include buttons **330a**, control sticks **330b**,

directional d-pad **330c**, and shoulder buttons **330d**. The housing **310** is shaped to be held in both hands of the user **42** such that the user's **10** thumbs are positioned to operate controls located on the front of the user device **300** (e.g., the control sticks **330b**) and the user's **10** index fingers are positioned to operate controls located on the top edge of the user device **300** (e.g., shoulder buttons **330d**). The user device **300** may also include one or more additional controls located on the back of the user device **300** in a position to be operated by the user's **10** other fingers. The user device **300** may include a dedicated device/controller for controlling the robot, or may include an off-the-shelf computing device that may execute one or more applications **324** for use in controlling the robot **10**. For instance, the user device **300** may include a mobile computing device, such as a smart phone or tablet that may execute one or more applications **324** for controlling the robot **10**.

[0045] The display (e.g., screen) **320** shows the user **42** an image or scene **322** of the environment **8** based on the sensor data **17** captured by the robot **10**. In some examples, the screen **320** presents the scene **322** as a video feed.

[0046] The user device **300** further includes, inside the housing **310**, a processing device **340**, a storage device **350** (e.g., memory device), and a communication interface **360**. In the example shown, the processing device **340** executes a graphical user interface (GUI) **321** for display on the screen **320** to present the scene **322** of the environment **8** for the user **42** to view. The user device **300** may include additional components not depicted in FIG. **3**. The processing device **340** can include one or more processors that execute computer-executable instructions and associated memory (e.g., RAM and/or ROM) that stores the computer-executable instructions. In implementations where the processing device **340** includes more than one processor, the processors can execute in a distributed or individual manner. The processing device **340** can execute an operating system **342** one or more applications for navigating the robot **10** in the environment **8**. In some examples, the processing device **340** executes other applications **344**, which may be native or non-native to the user device **300**, such as a web browser, all of which can be implemented as computer-readable instructions. In some examples, the processing device **340** executes at least a portion of the online authoring system **100**.

[0047] The storage device **350** can include one or more computer-readable mediums (e.g., hard disk drives, solid state memory drives, and/or flash memory drives). The storage device **350** can store any suitable data that is utilized by the operating system **342**, and the other applications **344**. The communication interface **360** includes one or more devices that are configured to communicate with the robot **10** in the environment **8**. In some implementations, the communication interface **360** is configured to communicate with the robot **10** by way of a network. The communication interface **360** can include one or more transceivers for performing wired or wireless communication. Examples of the communication interface **360** can include, but are not limited to, a transceiver configured to perform communications using the IEEE 802.11 wireless standard, an Ethernet port, a wireless transmitter, and a universal serial bus (USB) port.

[0048] Thus, the robot **10** collects information (e.g., sensor data **17**) about the environment **8** surrounding the robot **10** as well as information associated with operating in the environment **8**. The robot **10** may send some or all of this information to the user device **300**, including an image of the environment **8** captured by the robot **10**. The user device **300** then displays the image to the user **42** so that the user **42** can view the environment **8** that the robot **10** is operating in, which in this example is the interior of a warehouse. In some examples, the user **42** views the environment **8** from the perspective of the robot **10**. Such a "first person" experience may be beneficial to certain applications of the online authoring system **100**. The user device **300** displays the image of the environment **8** on the graphical user interface **321** and the user **42** may provide a navigation input indicating selection of a location within the image that corresponds to a physical location (e.g., a target location **T**) in the environment **8** where the user **42** wants the robot **10** to navigate. In some examples, the navigation input indicating selection of the location within the image is sent from the user device **300** to the robot **10** as a point in world space corresponding to an origin the scene of the

environment **8** is rendered from, as well as a direction in world space corresponding to the location within the image selected by the user **42**. For instance, the user device **300** may execute the GUI **321** for displaying the image of the environment **8** on the screen **320** of the user device **300** and the user **42** may provide the navigation input by touching the location within the image to navigate the robot **10** to the target location **T**. In additional examples, the user **42** provides the navigation input via a physical button, mouse, joystick, or any other input mechanism. As used herein, the navigation input refers to an input indication selection of the location with the image that corresponds to the target location **T** (e.g., physical location).

[0049] Referring now to FIG. **4**, in some implementations, the GUI **321** of the user device **300** presenting a playbook **410** of one or more actions **420**, **420a-n** for the robot **10** to perform at a corresponding location of at least one of the one or more target locations **52** in the environment. The GUI **321** may display an action selection window **422** presenting one or more of the actions **420**, **420a-c** that are available for selection by the user **42** for the robot **10** to perform. That is, when the current position of the robot **10** reaches the target location **52** while generating the environmental map **114**, the GUI **321** may receive a user input indication indicating a selection of a respective action **420** from the playbook **410** that the user **42** wants the robot **10** to perform at the target location **52**. In response to receiving the user input indication (e.g., cursor) **440** indicating selection of the respective action **420**, the user device **300** may issue an action recording request **450** (FIG. **1**) that causes the action recorder **120** to record the respective action **420** for the robot **10** to perform at the corresponding location of the target location **52**. The user device **300** may display the GUI **321** on the screen **320** of the user device **300**.

[0050] In the example shown, the exemplary actions **420** available for selection in the action selection window **422** include evaluating a door **50** (e.g., determining whether a door is open or closed), opening a door **420b**, and closing a door **420c**. In some examples, actions **420** are sorted into categories **430**, **430a-n**. For example, categories **430** include a door action category **430a**, a stair action category **430b**, and a lever action category **430c**. The user **42** may interact with any category **430** (e.g., via a cursor **440**, via touch, etc.) to display the corresponding actions **420** in that category **430**. For instance, a user input indication **440** indicating selection of the stair actions category **430** may cause the action selection window **422** to present one or more actions that relate to stairs in the environment **8**. Thus, the user **42** may navigate the robot **10** via the user device **300** to a target location **52** and then select the appropriate action **420** from the playbook **410** for that target location **52**.

[0051] Referring back to FIG. **1**, in some examples, prior to recording the respective action **420** for the robot **10** to perform, the action recorder **120** receives the action recording request **450** from the user device **300** when the current position of the robot **10** within the environmental map **114** reaches the corresponding target location **52**. The action recording request **450** requests the action recorder **120** to record the respective action **420** selected from the playbook **410** (FIG. **4**) for the robot **10** to perform at the corresponding target location **52**. That is, the user **42**, via the user device **300**, may navigate the robot **10** to the corresponding target location **52** prior to sending the action recording request **450** to the robot **10**. For example, when the intended action **420** includes interacting with the door **50**, the robot **10** may first navigate to the target location **52** (FIG. **1**) in front of the door **50**.

[0052] Optionally, recording the respective action **420** for the robot **10** to perform at the corresponding target location **52** includes the online authoring system **100** associating the current position of the robot **10** within the environmental map **114** with the respective action **420** for the robot **10** to perform at the corresponding target location **52** and storing (e.g., in memory hardware **38**) the association of the current position of the robot **10** within the environmental map **114** with the respective action **420** for the robot **10** to perform at the corresponding target location **52**. That is, the current location of the robot **10** may be the same as the target location **52**. Alternatively, the user **42** may indicate the target location **52** via the environmental map **114** or sensor data **17**

(displayed via the user device **300**) and the robot **10** may reference the target location **52** to the current location of the robot **10**. For example, in the example of FIG. **4**, the user **42** may select/reference the target location **52**, T depicted in the scene **322** ahead of the current position of the robot **10** and associate the respective action **420** with the target location **52**, T so that when the robot **10** navigates to the target location **52** during a future mission, i.e., after generating the map **114** and recording the respective action **420**, the robot **10** will perform the respective action **420**. [0053] Conventional techniques for recording actions **420** for future missions involve a user manually creating and editing a behavior tree. For example, FIG. **5** shows a conventional user interface **500** displaying a behavior tree **122** that includes a sequence of behaviors **510** that have been manually selected by a user. This process of manually selecting each behavior in the sequence of behavior assemble an action for a robot to perform is often tedious, slow, and prone to errors as an error in a portion of the sequence will propagate through all subsequent behaviors in the sequence.

[0054] In contrast to this conventional technique, implementations herein include the online authoring system **100** automatically generating the behavior tree **122** in response to the action(s) **420** selected by the user **42** from the playbook **410**. That is, in some examples, the playbook **410** of actions **420** associated with the corresponding target location **52** includes a collection of prebuilt behavior tree portions **620**, **620a-n** associated with the corresponding target location **52**. Each prebuilt behavior tree portion **620** includes a plurality of nodes **622**. Each node may be a root node **622a**, a control flow node **622b**, or an execution node **622c**. The behavior tree portion **620** begins at the root node **622** and each control flow node **424** (e.g., a selector node or sequence node) controls subtasks below it. The execution nodes **426** each perform a simple task such as standing, stopping, or taking a picture. These nodes **622**, when combined with other tasks allow the robot to perform a complex action **420** at the corresponding target location **52**. The action recorder **120** may generate the behavior tree **122** by combining two or more prebuilt behavior tree portions **620**. Each prebuilt behavior tree portion **620** of the combined two or more prebuilt behavior tree portions **620** may be associated with a different corresponding target location **52** and selected from the corresponding collection of prebuilt behavior tree portions **620** associated with the corresponding target location **52**.

[0055] FIGS. **6A** and **6B** provide sections **122a**, **122b** of a behavior tree **122** automatically generated by the online authoring system **100** to perform an “Autowalk” which includes navigating the robot **10** to a target location **52** within the environmental map **114** and taking a picture. FIG. **6A** illustrates a first prebuilt portion **620a** (for an “Autowalk” element) that has the children nodes **622** collapsed for illustration purposes. A second portion **620b** includes an “Autowalk” element with each of the children nodes **622** illustrated. The second portion **620b** also includes the nodes **622** illustrated in FIG. **6B**. These exemplary nodes provide the execution sequence for the robot **10** to, for example, navigate to the target location **52**, stand, stop, take a picture, etc. In some implementations, the user **42** may edit or modify the behavior tree **122** after generation. For example, the user **42** may adjust the structure of the behavior tree **122** or the contents of each portion **620**.

[0056] In some examples, the action recorder **120** determines a corresponding orientation for the robot **10** for each respective action **420** recorded for the robot **10** to perform. For example, when the user **42** (via the user device **300**) explicitly places the robot in a specific orientation while traversing and mapping the environment **8**, the online authoring system **100** may automatically generate the behavior tree **122** with the same specific orientation.

[0057] Optionally, after generating the environmental map **114** and the behavior tree **122**, and while the robot **10** navigates through the environment **8** during the future mission, the online authoring system **100** determines, using the environmental map **114** and the behavior tree **122**, when the current position of the robot **10** within the environmental map **114** reaches one of the one or more target locations **52** within the environment **8** about the robot **10**. When the current position of the

robot **10** within the environmental map **114** reaches the one of the one or more target locations **52**, the online authoring system may control the robot **10** to perform the respective action **420** at the corresponding target location **52**.

[0058] With reference to FIGS. **1**, **7A**, and **7B**, in some implementations, in response to receiving the action recording request **450** from the user device **300** to record a respective action **420** for the robot **10** to perform at a particular target location **52**, the online authoring system **100** obtains additional information **712** relevant to the respective action **420**. Recording the respective action **420** for the robot **10** to perform at the corresponding target location **52** may also include recording the additional information **712** relevant to the respective action **420** for the robot **10** to perform at the corresponding target location **52**. For instance, the additional information **712** may include additional data the online authorizing system **100** requires for recording so that the respective action **420** can be performed by the robot **10** during a future mission. FIGS. **7A** and **7B** show the GUI **321** displayed on the screen **320** of the user device **300** presenting a scene **322** of the environment **8** in front of the robot **10** for the user **42** to view when the robot **10** navigates to the target location **52** (FIG. **1**) in front of the door **50**. After selecting a relevant door action **420** (FIG. **4**) from the playbook of actions **410** (FIG. **4**), the online authoring system **100** may prompt **710** the user device **300** to obtain the additional information **712** relevant to the door action **420**.

Specifically, FIG. **7A** shows the user device **300** displaying a first prompt **710a** received from the online authoring system **100** in the GUI **121** that requests/prompts the user **42** to “Please select the location of the door handle”. The user **42** may provide an input indication to the GUI **321** (e.g., using a cursor **720** or via touch with a finger or stylus) indicating selection of a location at or near the door handle **730**, thereby causing the user device **300** to communicate the location of the door handle **730** to online authoring system **100** of the robot **10**. The robot may store the location of the door handle **730**.

[0059] Referring now to FIG. **7B**, in some examples, the online authoring system **100** issues a second prompt **710b** prompting the user device **300** to obtain even more additional information **712** relevant to the door action **420**. In the example shown, the user device **300** displays the second prompt **710b** received from the system **100** in the GUI **121** that requests/prompts the user **42** to “Please select the location of the door hinges”. Similar to selection of the door handle **730**, the user **42** may provide another input indication to the GUI **321** (e.g., with cursor **720** or via touch with a finger, stylus, etc.) indicating selection of a location at or near each of the door hinges **740a**, **740b** displayed within the scene **322**, thereby causing the user device **300** to communicate the location of the hinges **740a**, **740b** to the online authoring system **100** of the robot **10**. The robot **10** may store the location of the door hinges **740a**, **740b**. Thus, prompting the user **42** to provide the additional information **712** during mapping and while the user **42** is present allows the online authoring system **100** to gather/obtain accurate data from the user **42** and the environment **8** without the need to store large quantities of sensor data **17**.

[0060] FIG. **8** is a flowchart of an exemplary arrangement of operations for a method **800** of online authoring of robot autonomy applications. The method **800**, at operation **802**, includes receiving, at data processing hardware **36** of a robot **10**, from at least one sensor **31** in communication with the data processing hardware **36**, sensor data **17** of an environment **8** about the robot **10** while the robot **10** traverses through the environment **8**. At operation **804**, the method **800** includes generating, by the data processing hardware **36**, an environmental map **114** representative of the environment **8** about the robot **10** based on the received sensor data **17** of the environment **8** about the robot **10** while the robot **10** traverses through the environment **8**.

[0061] While generating the environmental map **114**, the method **800** includes, at operation **806**, localizing, by the data processing hardware **36**, a current position of the robot **10** within the environmental map **114**. At operation **808**, while generating the environmental map **114**, the method **800** includes, at each corresponding target location **52** of one or more target locations **52** within the environment **8** about the robot **10**, recording, by the data processing hardware **36**, a

respective action **420** for the robot **10** to perform. Still while generating the environmental map **114**, the method **800** includes, at operation **810**, generating, by the data processing hardware **36**, a behavior tree **122** for navigating the robot **10** to each target location **52** and controlling the robot **10** to perform the respective action **420** at each corresponding target location **52** of the one or more target locations **52** within the environment **8** during a future mission when the current position of the robot **10** within the environmental map **114** reaches the corresponding target location **52**.

[0062] FIG. **9** is schematic view of an example computing device **900** that may be used to implement the systems and methods described in this document. The computing device **900** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

[0063] The computing device **900** includes a processor **910**, memory **920**, a storage device **930**, a high-speed interface/controller **940** connecting to the memory **920** and high-speed expansion ports **950**, and a low speed interface/controller **960** connecting to a low speed bus **970** and a storage device **930**. Each of the components **910**, **920**, **930**, **940**, **950**, and **960**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **910** can process instructions for execution within the computing device **900**, including instructions stored in the memory **920** or on the storage device **930** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **980** coupled to high speed interface **940**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **900** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0064] The memory **920** stores information non-transitorily within the computing device **900**. The memory **920** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **920** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **900**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

[0065] The storage device **930** is capable of providing mass storage for the computing device **900**. In some implementations, the storage device **930** is a computer-readable medium. In various different implementations, the storage device **930** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **920**, the storage device **930**, or memory on processor **910**.

[0066] The high speed controller **940** manages bandwidth-intensive operations for the computing device **900**, while the low speed controller **960** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **940** is coupled to the memory **920**, the display **980** (e.g., through a graphics processor or

accelerator), and to the high-speed expansion ports **950**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **960** is coupled to the storage device **930** and a low-speed expansion port **990**. The low-speed expansion port **990**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

[0067] The computing device **900** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **900a** or multiple times in a group of such servers **900a**, as a laptop computer **900b**, or as part of a rack server system **900c**.

[0068] Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

[0069] A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

[0070] These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0071] The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The

processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

[0072] To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

[0073] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A method comprising: receiving, at data processing hardware of a robot, sensor data associated with an environment of the robot and corresponding to a first traversal of the environment; identifying, by the data processing hardware, first data indicative of a selection of a first action from a first plurality of actions, wherein the first plurality of actions is based on the sensor data, and wherein the first plurality of actions is associated with a first location in the environment; identifying, by the data processing hardware, second data indicative of a manner of performing the first action; generating, by the data processing hardware, a mission, wherein the mission indicates the robot is to navigate to the first location, to perform the first action at the first location based on the first data and the second data, and to navigate to a second location; and instructing, by the data processing hardware, the robot to autonomously perform the mission, wherein performance of the mission corresponds to a second traversal of the environment.
2. The method of claim 1, wherein the second data indicates an orientation associated with performance of the first action.
3. The method of claim 1, wherein the second data indicates a third location of one or more objects associated with performance of the first action.
4. The method of claim 1, wherein the first action comprises one or more of an action to interact with a door, an action to traverse a stair, or an action to interact with a lever.
5. The method of claim 1, wherein the first plurality of actions corresponds to a first category of actions of a plurality of categories of actions.
6. The method of claim 1, wherein the first plurality of actions comprises actions to interact with or evaluate an object in the environment.
7. The method of claim 1, further comprising: identifying the first plurality of actions based on the first traversal of the environment, wherein the first plurality of actions is linked to the first location; generating an environmental map representative of the environment based on the received sensor data; and providing, to a user computing device, data indicative of the environmental map and the first plurality of actions.
8. The method of claim 1, wherein a second plurality of actions is linked to the second location, wherein the second plurality of actions is different relative to the first plurality of actions, and wherein the mission further indicates the robot is to perform a second action, selected from the second plurality of actions, at the second location.
9. The method of claim 1, wherein instructing the robot to autonomously perform the mission

comprises: generating computer-executable instructions; and providing, to the robot, the computer-executable instructions, the method further comprising: executing, by the robot, the computer-executable instructions.

10. The method of claim 1, further comprising: generating a behavior tree based on the first action, wherein the behavior tree links the first location to the first action, and wherein the mission is based on the behavior tree.

11. The method of claim 1, wherein identifying the first data comprises: receiving, from a user computing device, an action recording request requesting recordation of the first action for the robot to perform at the first location, the method further comprising: automatically linking the first action to the first location in response to receiving the action recording request.

12. The method of claim 1, wherein the first traversal of the environment and the second traversal of the environment are a traversal of a same portion of the environment.

13. The method of claim 1, further comprising: receiving, from a user computing device, navigation instructions, wherein the navigation instructions instruct the robot to navigate to the first location, wherein the mission is based on the navigation instructions.

14. The method of claim 1, further comprising: providing, to a user computing device, second sensor data associated with the environment, wherein receiving the first data is based on providing the second sensor data.

15. The method of claim 1, further comprising: receiving third data indicative of the first location, wherein generating the mission is further based on receiving the third data.

16. A system comprising: data processing hardware; and memory in communication with the data processing hardware, the memory storing instructions that when executed on the data processing hardware cause the data processing hardware to: receive sensor data associated with an environment of an robot and corresponding to a first traversal of the environment; identify first data indicative of a selection of a first action from a first plurality of actions, wherein the first plurality of actions is based on the sensor data, and wherein the first plurality of actions is associated with a first location in the environment; identify second data indicative of a manner of performing the first action; generate a mission, wherein the mission indicates the robot is to navigate to the first location, to perform the first action at the first location based on the first data and the second data, and to navigate to a second location; and instruct the robot to autonomously perform the mission, wherein performance of the mission corresponds to a second traversal of the environment.

17. The system of claim 16, wherein the first traversal of the environment comprises a first traversal of the environment by one or more robots.

18. The system of claim 16, wherein generating the mission is based on identifying the first data and identifying the second data.

19. A robot comprising: at least two legs; data processing hardware; and memory in communication with the data processing hardware, the memory storing instructions that when executed on the data processing hardware cause the data processing hardware to: receive sensor data associated with an environment of the robot and corresponding to a first traversal of the environment; identify first data indicative of a selection of a first action from a first plurality of actions, wherein the first plurality of actions is based on the sensor data, and wherein the first plurality of actions is associated with a first location in the environment; identify second data indicative of a manner of performing the first action; generate a mission, wherein the mission indicates the robot is to navigate to the first location, to perform the first action at the first location based on the first data and the second data, and to navigate to a second location; and instruct the robot to autonomously perform the mission, wherein performance of the mission corresponds to a second traversal of the environment.

20. The robot of claim 19, wherein the mission further indicates the robot is to navigate from a location of the robot to the first location, to determine that the location of the robot corresponds to the first location, to perform the first action at the first location further based on determining that

the location of the robot corresponds to the first location, and to navigate to a second location based on performance of the first action.
